PULLBACK OF A NORMAL CROSSINGS DIVISOR BY A SMOOTH MORPHISM IS A NORMAL CROSSINGS DIVISOR

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 02OO: sources/stacks-chow/tags.txt, source lines 3523–3523. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0BSF: sources/stacks-chow/tags.txt, source lines 14803–14803. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GO: sources/stacks-chow/tags.txt, source lines 3235–3235. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CBP: sources/stacks-chow/tags.txt, source lines 15479–15479. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $D \subset X$ be a normal crossings divisor. If $f : Y \to X$ is a smooth morphism of schemes, then the pullback f^*D is a normal crossings divisor on Y.*

Proof. As f is flat the pullback is defined by Divisors, Lemma 02OO hence the statement makes sense. Let $q \in f^*D$ map to $p \in D$. Choose an étale morphism $U \to X$ whose image contains p such that $D \times_X U \subset U$ is a strict normal crossings divisor as in Definition 0BSF. Set $V = Y \times_X U$. Then $V \to Y$ is étale as a base change of $U \to X$ (Morphisms, Lemma 02GO) and the pullback $D \times_X V$ is a strict normal crossings divisor on V by Lemma 0CBP. Thus we have checked the condition of Definition 0BSF for $q \in f^*D$ and we conclude. □

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